

“In the x86 space, Intel Atom has opened a new market for SFF designs”



Headquartered in Germany, Kontron is a leading embedded computing technology provider. With more than 30 per cent of its employees in R&D, Kontron creates many of the standards that drive embedded computing platforms.

In India, the company has expansion plans for design services and system integration in selected market verticals for local customers. Pradip Kulkarni, managing director, Kontron Technology India, speaks to Uma Gupta of EFY on the trends in small-form-factor embedded computing and Kontron's activities in India

PRADIP KULKARNI
MANAGING DIRECTOR, KONTRON TECHNOLOGY INDIA

How are highly integrated small-form factors (SFFs) redefining different application areas?

We can spot the SFF trend everywhere. It is generic for all application areas being a purely technology-driven improvement, which is applied by all silicon vendors. This is because all applications benefit from reduced power consumption and lower heat dissipation. Also, every application benefits from a reduced bill of material due to more efficient and simplified system designs. Of course, some application areas benefit in particular, e.g., mobile devices, because it enables increasing computing and graphics power on even smaller footprints.

What are the chipsets and processors at the heart of these platforms?

In line with the different application areas addressed, there is also differentiation between several processors and chipsets. In the x86 space it is definitely the Intel Atom processor technology that has opened a new market for SFF designs. Not to forget the field-programmable growth array (FPGA) implementation of the Intel Atom 6x5 processors, which is most interesting for highly integrated platforms.

But AMD has also made a significant contribution to the SFF area with its so-called accelerated AMD embedded G-Series processors, which unite the CPU together with a GPU on a single die. The latest ARM processor platforms from NVIDIA as well as from Texas Instruments are catching up

on the x86 technology with respect to the performance of ultra-low-power applications, and are also good examples of the SFF trend towards smart embedded devices with attractive graphical user interfaces.

How easy is to integrate these for desired applications?

x86 technologies are widely used and embrace a huge ecosystem in terms of standard platforms, inputs/outputs and software support. Things are different for the ARM sector because here usually one receives only dedicated support for dedicated devices. If something more sophisticated or tailored to the customer's needs is required, one basically has to take care of this himself because there are no standardised procedures.

This has changed with Kontron launching the new SFF computer-on-modules standard – a low-power embedded architecture platform for ARM and system-on-chip (SoC) designs along with extensive software services. The new component object model (COM) standard features two SFFs designed for the future ultra-low-power, off-the-shelf platforms, including embedded handheld devices and rugged tablets, as well as box PCs and human-machine interface systems.

By extending the COM usage model to reduced instruction set computing (RISC) architectures with scalable, modular and out-of-the-box solutions, this module standard bridges the gap between current proprietary industrial of-